

Power Amplifier Modeling Using Memory Polynomials with Delay-dependent Polynomial Orders

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The Evolution of Wireless Systems

Each generation demands more from the transmitter

- Higher data rates and spectral efficiency
- Wider signal bandwidths
- Complex modulation schemes with high peak-to-average power ratio (PAPR)
- Lower energy consumption and more efficient use of hardware resources

The power amplifier must handle increasingly demanding signals while preserving efficiency and signal quality.

Evolution of wireless systems



Higher data rates + wider bandwidths



High-PAPR signals + efficiency demands



Greater stress on the PA

Why PA Modeling Matters

- RF power amplifiers (PAs) are most efficient near saturation
- Near saturation, nonlinearities create in-band distortion and spectral regrowth
- Broadband signals also expose electrical and thermal memory effects
- Digital pre-distortion (DPD) needs an accurate behavioral model of the PA

Engineering challenge: retain modeling accuracy while reducing the cost of a real-time digital implementation.

Efficiency → saturation



Nonlinearity + memory



DPD model required

The Limitation of the Classic MP

Classic Memory Polynomial

$$y(n) = \sum_{m=0}^M \sum_{p=1}^P h_{p,m} x(n-m) |x(n-m)|^{p-1}$$

- Same maximum order P at every delay
- Linear in the coefficients
- Accurate and easy to identify by least squares

But memory branches are not equally relevant

- The current sample usually contains the strongest nonlinear contribution
- Older samples often contribute less
- A uniform order can add weak terms, operations, and registers

Research question: can complexity follow the relevance of each delay?

Delay-dependent Polynomial Orders

$$y(n) = \sum_{m=0}^M \sum_{p=1}^{P_m} h_{p,m} x(n-m) |x(n-m)|^{p-1}$$

- Each delay m receives its own maximum order P_m
- The classic MP is recovered when all P_m are equal
- The model remains linear in its coefficients
- Coefficients are still estimated by least squares

Expected order profile

Current	Delay 1	Delay 2
High order	Medium	Low order
P_0	P_1	P_2

Working hypothesis: $P_0 \geq P_1 \geq P_2$
captures most of the useful behavior with fewer terms.

Experimental Method

Datasets

- LDMOS PA: 4,500 complex samples for extraction and 4,500 for validation
- Class-AB GaN HEMT PA
- 900 MHz carrier, 3GPP WCDMA signal
- 3.84 MHz bandwidth, sampled at 61.44 MS/s

Search space

- Memory depth: $M = 2$
- $P_0, P_1, P_2 \in \{1, 2, 3, 4, 5\}$
- 125 independently estimated configurations

Evaluation workflow

1. Build the regression matrix
2. Estimate coefficients by least squares
3. Validate on independent samples
4. Compare NMSE and coefficient count
5. Validate fixed-point VHDL and synthesize with Yosys

$$\text{NMSE} = 10 \log_{10} \left(\frac{\sum |e(n)|^2}{\sum |y_{\text{measured}}(n)|^2} \right)$$

More negative is better.

Sensitivity Is Concentrated at the Current Sample

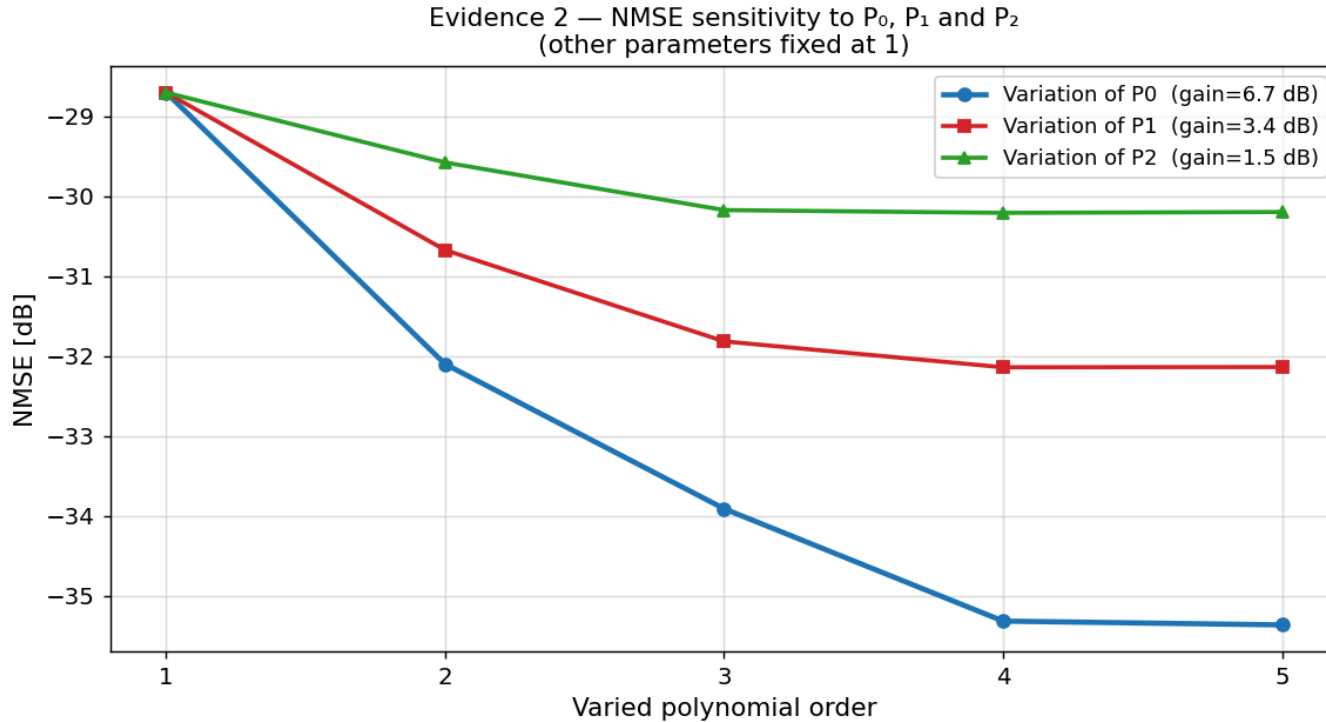


Figure 1: LDMOS NMSE sensitivity when each branch order varies from 1 to 5.

NMSE gain

Branch	Gain
P_0	6.66 dB
P_1	3.43 dB
P_2	1.49 dB

Complexity added to P_0 is substantially more valuable than complexity added to older delays.

Accuracy with Fewer Coefficients

Orders	Coefficients	GaN NMSE	LDMOS NMSE
(1, 1, 1)	3	−21.66 dB	−28.71 dB
(3, 2, 1)	6	−25.53 dB	−34.75 dB
(4, 3, 2)	9	−25.64 dB	−36.31 dB
(5, 3, 2)	10	−25.49 dB	−37.00 dB
(5, 5, 5)	15	−26.10 dB	−37.51 dB

(4, 3, 2) uses **40% fewer coefficients** and remains within 0.46 dB of the full GaN model.

(5, 3, 2) uses **33% fewer coefficients** and remains within 0.51 dB of the full LDMOS model.

The Improvement Is Also Spectral

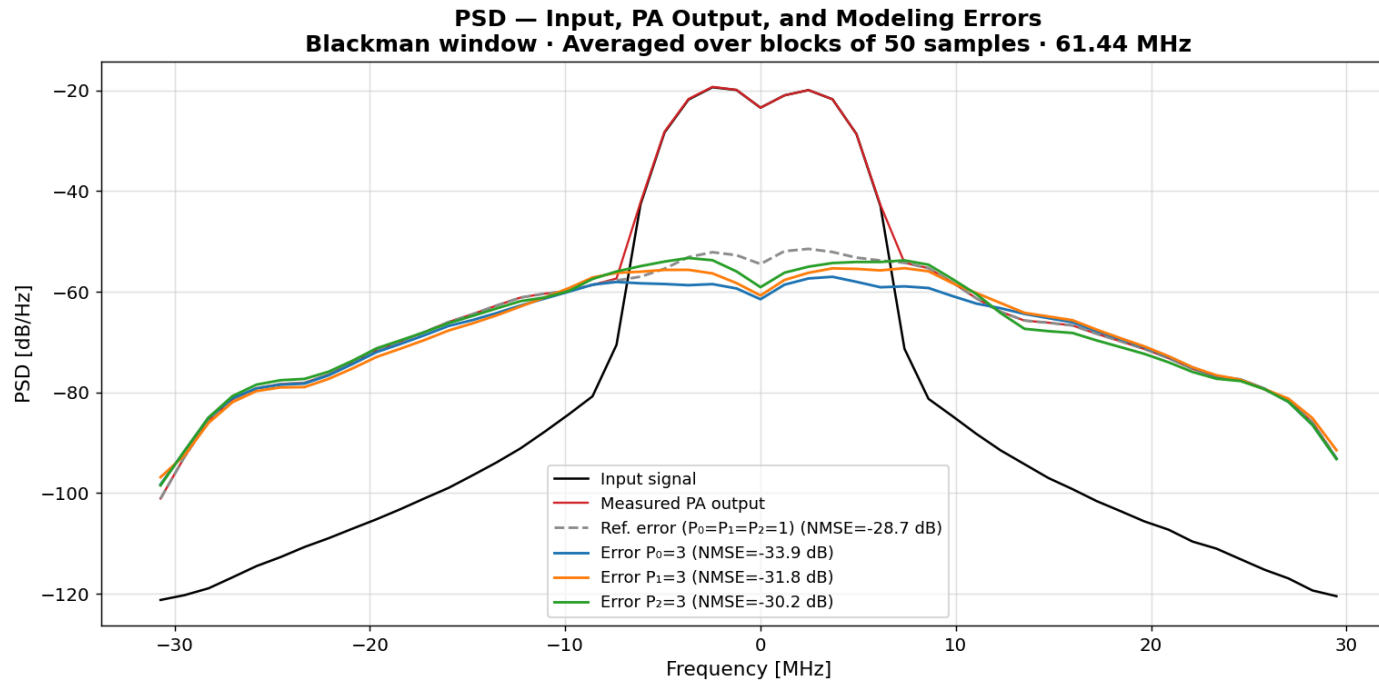


Figure 2: Measured spectrum and modeling-error spectra for selected configurations.

- Better time-domain NMSE is reflected in lower spectral error
- Higher order in the important branches lowers error near the main and adjacent bands
- Increasing older-delay orders yields comparatively modest improvement

This is the behavior needed for DPD-oriented modeling.

Hardware Cost Follows Structural Complexity

Architecture	Coef.	Cells	Registers
(5, 5, 5)	15	100%	100%
(5, 3, 2)	10	60%	60%
(4, 3, 2)	9	58%	58%

- Fixed-point VHDL matched the Python reference behavior
- GHDL was used for validation and Yosys for logic synthesis
- Truncation removes partial products and reduces adder depth and register pressure
- No target FPGA or place-and-route was used at this stage

The (5, 3, 2) model reduces the main synthesis resources by about **40%** while keeping LDMOS NMSE within **0.51 dB** of the full model.

Pareto Frontier: The Design Decision

- A Pareto-optimal model cannot improve accuracy without increasing complexity
- Configurations such as (4, 3, 2) and (5, 3, 2) occupy the useful trade-off region
- High orders at weak memory branches consume resources without proportional NMSE gains

Design rule: begin with a high current-sample order and progressively lower the orders across memory.

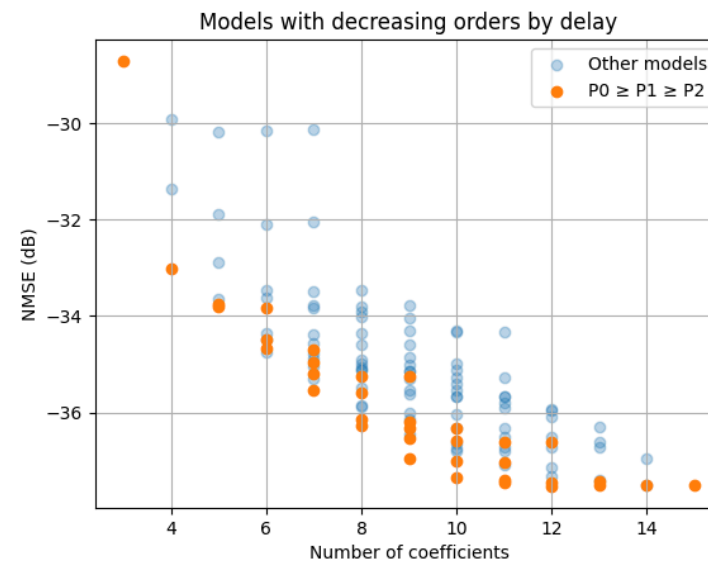


Figure 3: Complexity-performance trade-off across the candidate models.

Takeaways

What the evidence shows

- Nonlinear relevance decays across the analyzed memory delays
- Delay-dependent orders preserve most of the classic MP accuracy
- The trend is consistent across GaN and LDMOS datasets

Why it matters

- Fewer coefficients and arithmetic operations
- About 40% lower main synthesis-resource counts for selected models
- A practical path toward compact real-time DPD implementations

Main message: polynomial complexity should be allocated according to delay relevance, not uniformly.